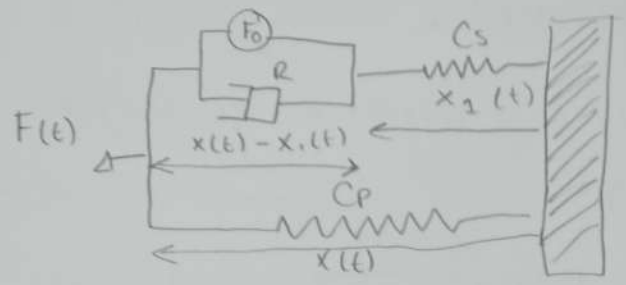
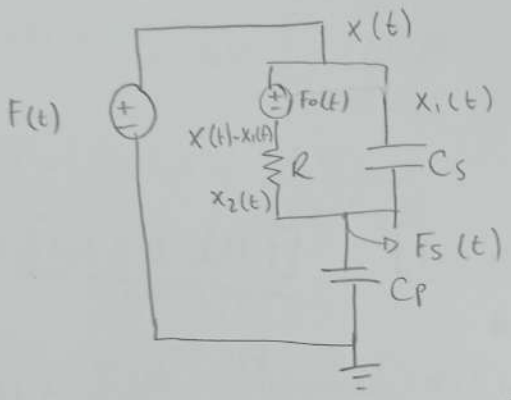
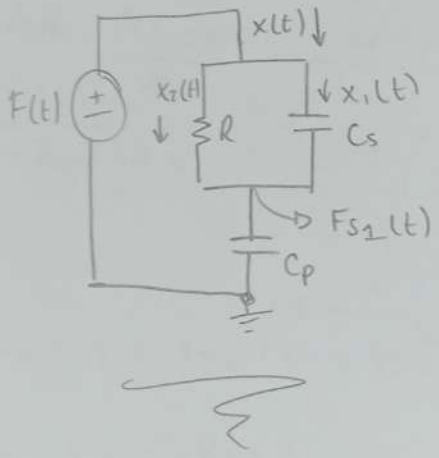
# 5.5 Sistema Musculoesquelético



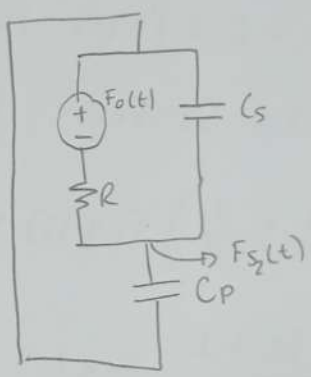




Superposición

(1)

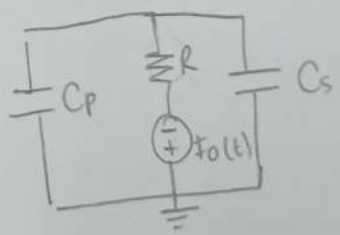



(2)



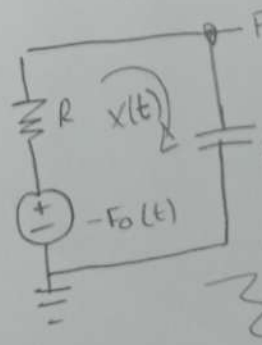

↓

(2')




↓

(2)




$F_o(t) = \alpha F(t)$

Circuito 2: Apagando $F(t)$

$$x(t) = R\,\frac{d[-Fo(t) - Fs_2(t)}{dt} + (C_s + C_p)\,\frac{d\,Fs_2(t)}{dt}$$

$$= R(-Fo(s)) - R(Fs_2(s) + C_s\,Fs_2(s) + C_p\,Fs_2(s)$$

---

$$-\alpha F(t) = Rx(t) + \frac{1}{C_p + C_s}\int x(t)\,dt$$

$$Fs_2(t) = \frac{1}{C_p + C_s}\int x(t)\,dt$$

$$-\alpha F(s) = RX(s) + \frac{X(s)}{(C_p + C_s)s}$$

$$Fs_2(s) = \frac{X(s)}{(C_p + C_s)s}$$

$$Fs(s) = -\frac{X(s)}{\alpha}\,\frac{R(C_p + C_s)s + 1}{(C_p + C_s)s}$$

$$\frac{Fs_2(s)}{Fs(s)} = \frac{\dfrac{X(s)}{(C_p + C_s)s}}{-\dfrac{X(s)}{\alpha}\,\dfrac{R(C_p + C_s) + 1}{(C_p + C_s)s}} = -\frac{\alpha}{R(C_p + C_s)s + 1}$$

$$Fs(s) = Fs_1(s) + Fs_2(s)$$

$$Fs(s) = \frac{C_s R s + 1}{R(C_p + C_s)s + 1}\,F(s) - \frac{\alpha}{R(C_p + C_s)s + 1} = \frac{\alpha\,C_s R s + 1}{R(C_p + C_s)s + 1}$$

$$\frac{Fs(s)}{F(s)} = \frac{C_s R s + 1}{R(C_p + C_s)s + 1}$$

Función de transferencia